



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to M.M. AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

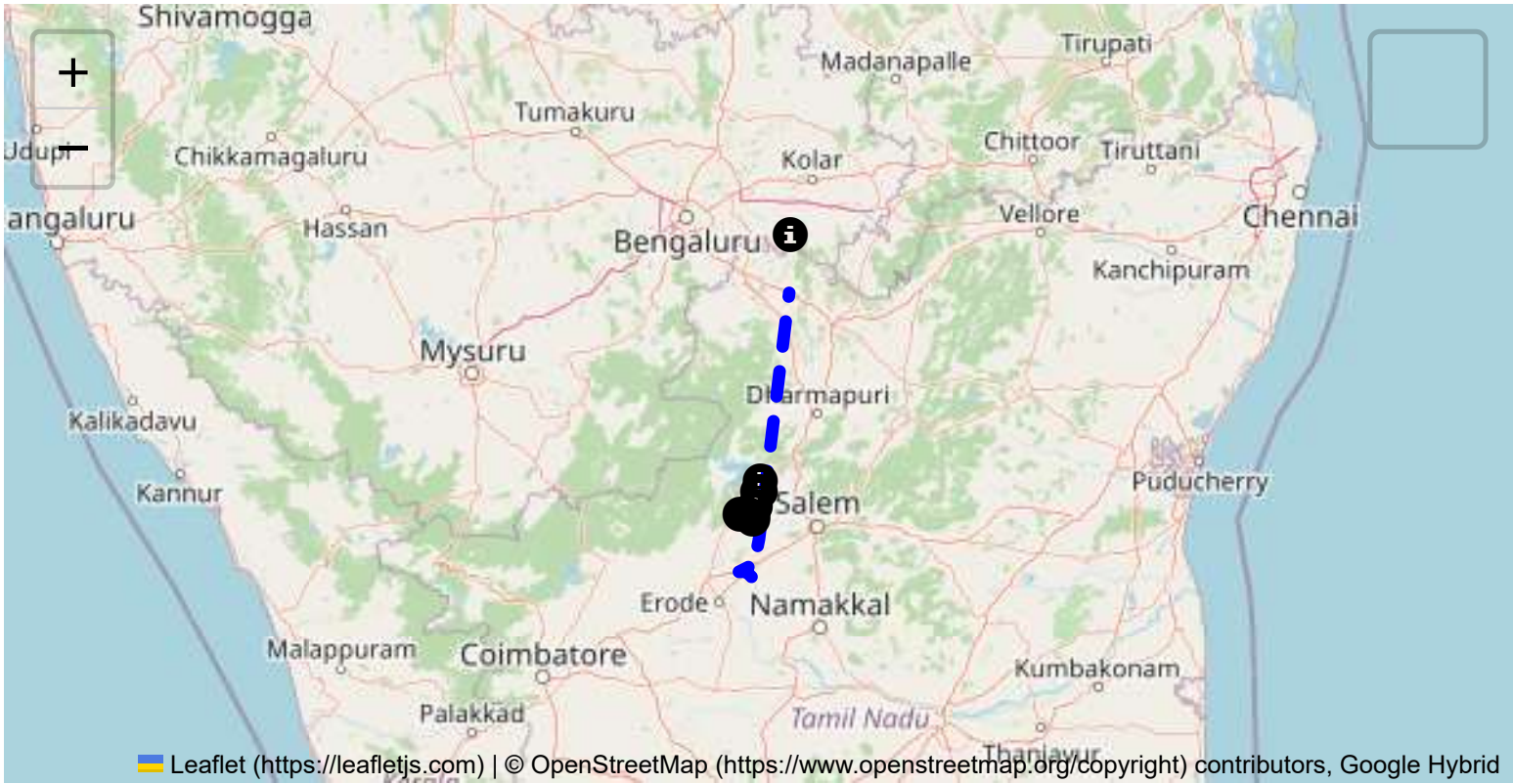
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

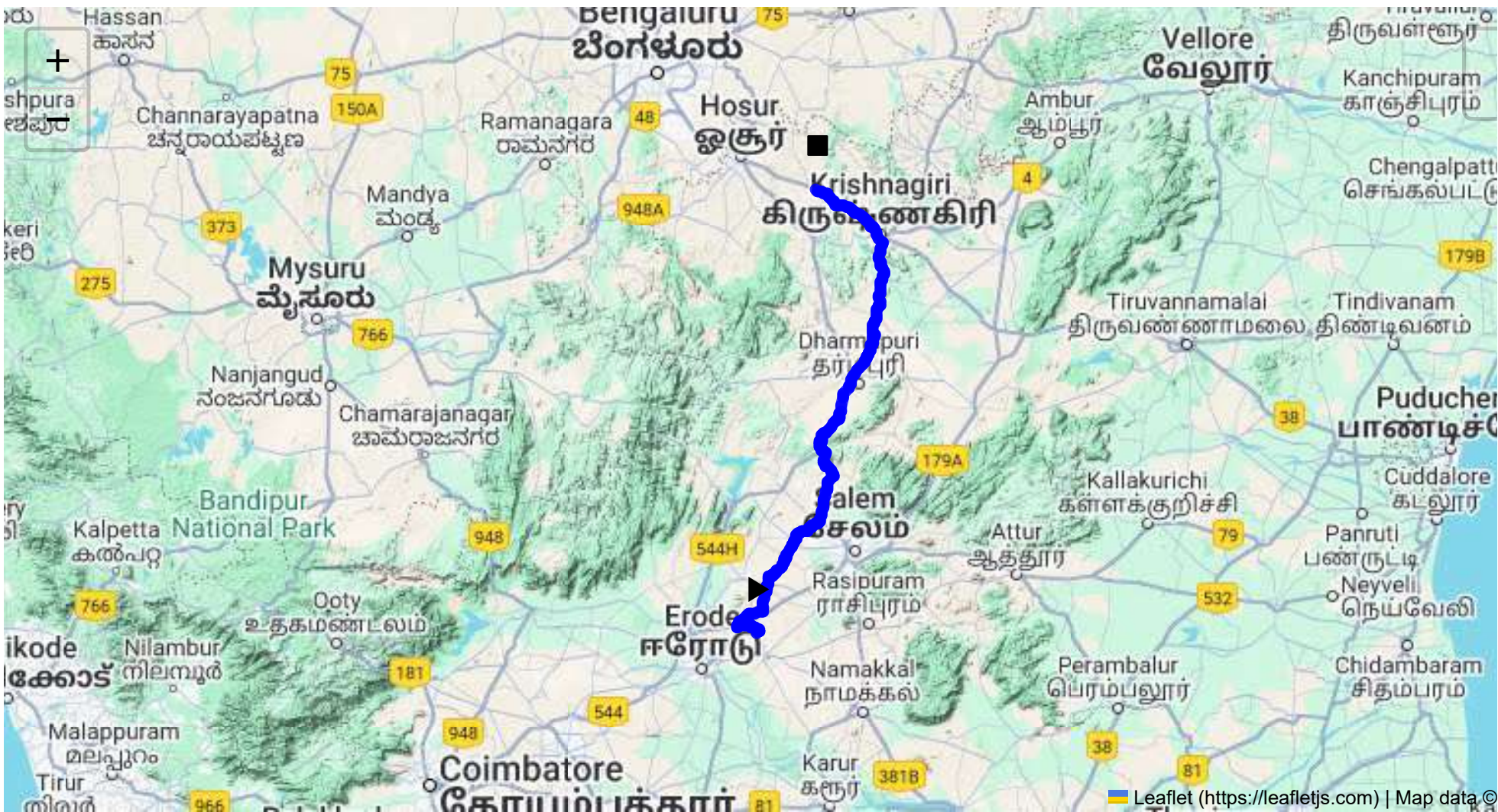
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 176.71 km
Estimated Duration: 3.4 hours
Adjusted Duration (Heavy Vehicle): 4.2 hours
Start: (11.4381, 77.8734)
End: (12.649813, 78.035268)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Gollapalli covers approximately 176.71 kilometers, primarily traversing through Tamil Nadu via major roads including the Sankari RS - Erode Privu Road and State Highway 86. It connects smaller towns such as Devannagoundanur and Konganapuram.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region experiences a tropical climate with hot, dry summers and moderate monsoons. The monsoon season (June to September) is marked by heavy rainfall that can lead to floods and poor visibility, increasing the risk of accidents due to slippery roads and water-logged sections. During the summer, high temperatures can lead to road surface damage.

3. Analysis of Traffic Patterns

Traffic congestion is typically higher near urban centers, especially during morning and evening hours. In particular, Bhavani Main Road in Sankari West and Padaiveedu could experience significant congestion during peak commuting times (8-10 AM and 5-8 PM). Truck traffic may slow down the journey, especially around junctions and market areas.

4. Assessment of Road Quality and Infrastructure

Overall, the road conditions are variable, with stretches of well-maintained highways interspersed with rural road segments that may have potholes and narrow lanes. Specific segments of SH 86 may require caution due to tight bends and uneven surfaces, particularly near Erumaipatti.

5. Suggestions for Alternative Routes for Emergencies

In case of emergencies, an alternative route via NH44 might be considered, although it may extend the travel time. This national highway offers better infrastructure and emergency services access.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials through Tamil Nadu requires adherence to national regulations, including permissions and special signage on vehicles. Careful route planning to avoid populated areas may be required as per local guidance.

7. Overview of Historical Incidents

Historically, there have been incidents involving heavy vehicles on Bhavani Main Road and SH 86 involving brake failures and road obstructions. Past incidents underline the importance of maintaining low speeds and vigilance.

8. Environmental Considerations and Sensitive Areas

Sensitive areas include school zones, religious sites, and marketplaces, all requiring reduced speed and heightened alertness. Wildlife crossings might be a concern in rural segments, hence speed restrictions may be enforced.

9. Analysis of Communication Coverage

The route generally has good cellular coverage; however, certain rural stretches, particularly near Koranampatti, may experience weak signals. Pre-planning communication hubs along the route can aid in maintaining contact.

10. Estimated Emergency Response Times

Urban areas such as Sankari and Erumaipatti have quicker emergency response, estimated around 20-30 minutes. Rural segments, however, may experience delays, extending response time beyond an hour.

12. Overall Summary of Risk Assessment

This route is moderate-to-high risk for heavy vehicles carrying hazardous materials due to varied road quality, potential for severe weather, and complex traffic dynamics in urban-linked areas. Pre-trip inspections, speed management, and weather monitoring are crucial for safety. Emergency protocols should be well understood and alternative routes planned in case of unforeseen closures or hazards.

In conclusion, thorough preparedness and adherence to regulations are necessary when navigating this route, ensuring safe transport of hazardous materials by considering traffic, weather, and infrastructure conditions comprehensively.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	Medium	11.45352, 77.85698	30 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	High	11.45571, 77.81702	15 KM/Hr
7	Turn	Medium	11.45654, 77.81706	30 KM/Hr
8	Turn	Medium	11.46916, 77.82431	30 KM/Hr
9	Blind Spot	Blind Spot	11.49395, 77.88537	10 KM/Hr
10	Turn	High	11.49465, 77.88488	15 KM/Hr
11	Turn	Medium	11.57649, 77.90610	30 KM/Hr
12	Turn	Medium	11.59214, 77.90656	30 KM/Hr
13	Turn	Medium	11.63833, 77.94818	30 KM/Hr
14	Turn	High	11.64243, 77.95198	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
15	Turn	Medium	11.64721, 77.95118	30 KM/Hr
16	Turn	Medium	11.67870, 77.96846	30 KM/Hr
17	Turn	High	11.70227, 77.97864	15 KM/Hr
18	Turn	Medium	11.71416, 77.99604	30 KM/Hr
19	Turn	Medium	11.71715, 77.99969	30 KM/Hr
20	Turn	Medium	11.71921, 78.01133	30 KM/Hr
21	Turn	Medium	11.72737, 78.02248	30 KM/Hr
22	Turn	High	11.73775, 78.03465	15 KM/Hr
23	Turn	High	11.73651, 78.03878	15 KM/Hr
24	Turn	High	11.74189, 78.04660	15 KM/Hr
25	Turn	Medium	11.78039, 78.05792	30 KM/Hr
26	Turn	Medium	11.86753, 78.08331	30 KM/Hr
27	Turn	Medium	11.86981, 78.08387	30 KM/Hr
28	Turn	Medium	11.88041, 78.08143	30 KM/Hr
29	Turn	Medium	11.91302, 78.05856	30 KM/Hr
30	Turn	Medium	11.92526, 78.06488	30 KM/Hr
31	Turn	Medium	11.93402, 78.06354	30 KM/Hr
32	Turn	Medium	11.94600, 78.04740	30 KM/Hr
33	Turn	Medium	11.95625, 78.04748	30 KM/Hr
34	Turn	High	11.95880, 78.04989	15 KM/Hr
35	Turn	High	11.96040, 78.04947	15 KM/Hr
36	Turn	High	11.96320, 78.05139	15 KM/Hr
37	Turn	Medium	11.96463, 78.05077	30 KM/Hr
38	Turn	High	11.96597, 78.04920	15 KM/Hr
39	Turn	Medium	11.98504, 78.06287	30 KM/Hr
40	Turn	Medium	12.17102, 78.16860	30 KM/Hr
41	Turn	High	12.24611, 78.19519	15 KM/Hr
42	Turn	Medium	12.25200, 78.20034	30 KM/Hr
43	Turn	High	12.40026, 78.21775	15 KM/Hr
44	Turn	Medium	12.40734, 78.22344	30 KM/Hr
45	Turn	Medium	12.50381, 78.22350	30 KM/Hr
46	Turn	Medium	12.54465, 78.20101	30 KM/Hr
47	Turn	Medium	12.59633, 78.13696	30 KM/Hr
48	Turn	Medium	12.63641, 78.07127	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
10	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
12	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
13	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
14	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
15	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
16	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
17	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
18	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
19	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
20	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
23	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
25	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
26	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
27	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
29	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
33	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium
34	clinic	Joe Hospital	12.5209601, 78.2150623	30 km/h	Medium
35	clinic	Ohm Prakash Clinic	12.5206321, 78.2159682	30 km/h	Medium
36	clinic	Srinivasa Clinic	12.520975, 78.215755	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
37	clinic	Dr. Anbalagan Clinic	12.5208366, 78.2160593	30 km/h	Medium
38	clinic	Shri Narayana Bone & Joint Clinic	12.523617, 78.210613	30 km/h	Medium
39	clinic	Shree Ramachandra ENT Centre	12.5290096, 78.2037359	30 km/h	Medium
40	hospital	Ga Heart Hospital	12.5287708, 78.2055336	30 km/h	Medium
41	clinic	Psychiatric Centre	12.529476, 78.205064	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
11	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
22	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
24	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
28	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
30	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
31	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium
32	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium
42	college	Government Medical College	12.582332, 78.1496363	30 km/h	Medium
43	school	Crescent Matriculation Higher Seconddary School	12.6003724, 78.132787	30 km/h	Medium
44	college	Jeeva College of Nursing	12.6067058, 78.1191944	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45764, 77.85219



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45571, 77.81702



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45654, 77.81706



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46916, 77.82431



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49395, 77.88537



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49465, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57649, 77.90610



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59214, 77.90656



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63833, 77.94818



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64243, 77.95198



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64721, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67870, 77.96846



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70227, 77.97864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71416, 77.99604



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71715, 77.99969



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71921, 78.01133



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72737, 78.02248



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73775, 78.03465



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03878



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.74189, 78.04660



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.78039, 78.05792



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86753, 78.08331



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86981, 78.08387



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88041, 78.08143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91302, 78.05856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92526, 78.06488



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93402, 78.06354



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.94600, 78.04740



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95625, 78.04748



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95880, 78.04989



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96040, 78.04947



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96320, 78.05139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96463, 78.05077



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96597, 78.04920



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.98504, 78.06287



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17102, 78.16860



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.24611, 78.19519



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.25200, 78.20034



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.40026, 78.21775



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.40734, 78.22344



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.50381, 78.22350



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.54465, 78.20101



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.59633, 78.13696



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.63641, 78.07127

